LawNet-Viz

A Web-based System to Visually Explore Networks of Law Article References

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OBJECTIVE

- Web-based tool for the modeling, analysis and visualization of law reference networks
- Support to legal research tasks for referred and semantically related articles
- Interactive, fluid and effective user experience

WHO IS LAWNET-VIZ FOR?

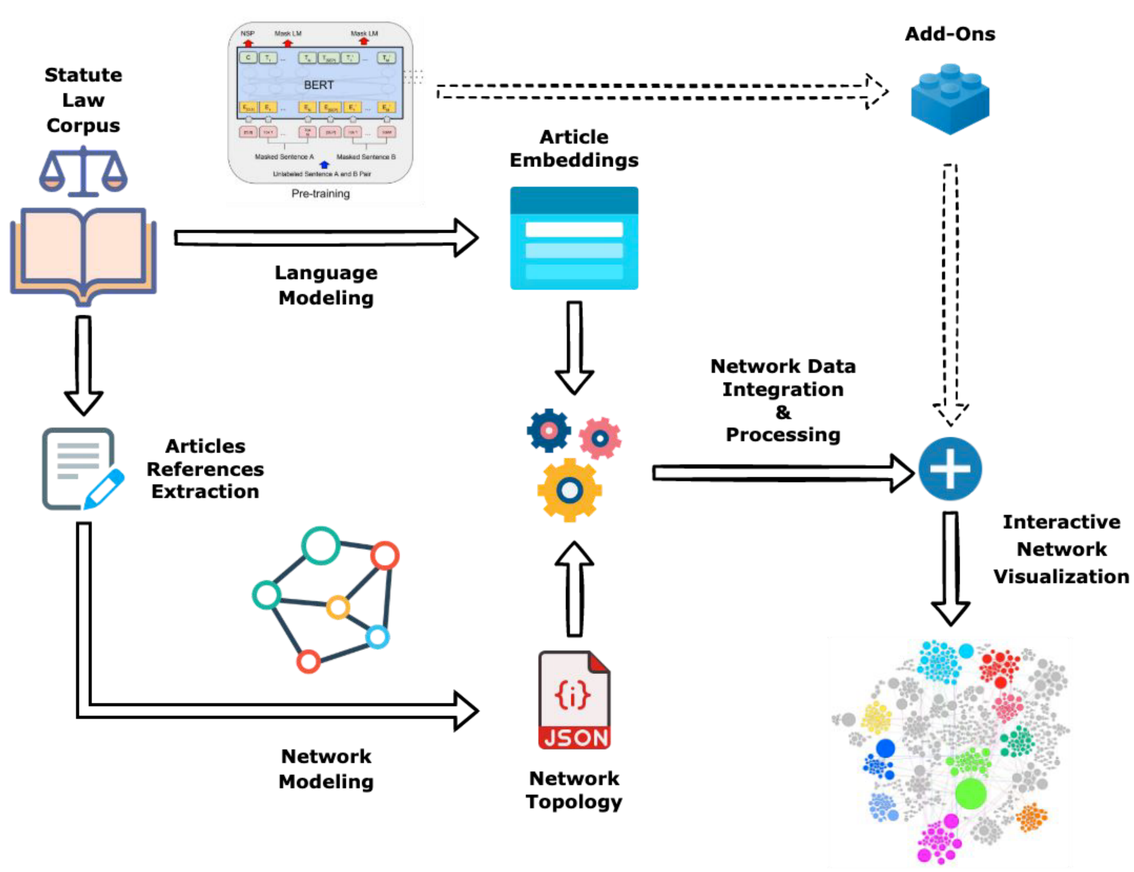
LEGAL EXPERTS

- Identify relevant statutes in a cost and time-effective manner
- Benefit in drafting new codes or abrogating existing ones, by inferring semantic relations between legal documents

LAYMEN/CITIZENS

- Legal aid to those who are not familiar with the legal domain
- Serve their search and consultation needs

WORKFLOW



NODE AND EDGE FEATURES

Network Topology
Overall number of articles and references in the network, depending on the filtering threshold

Article Metadata
Article ID, Article title, Article text, Book ID, Cluster ID, Layout coordinates, color, size

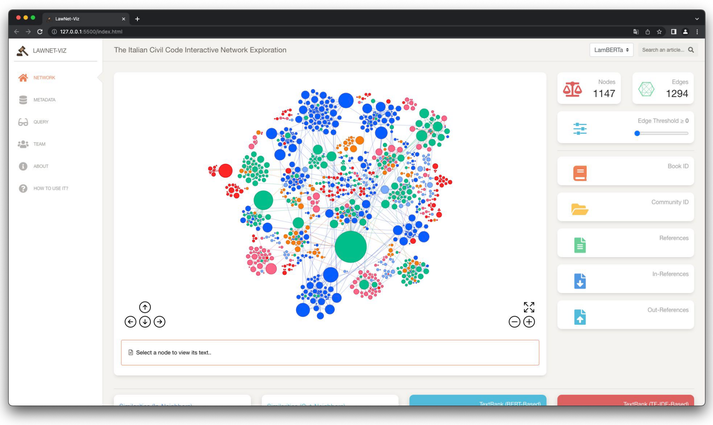
Article References
Overall, incoming and outgoing references involving a target article

Centrality Scores
Betweenness, closeness, PageRank, Eigenvector centrality, and two variants of TextRank

Similarity Weights
Cosine similarity values of the vectorial representations (e.g., TF-IDF, BERT, etc.) of the articles

INTERACTION MODES

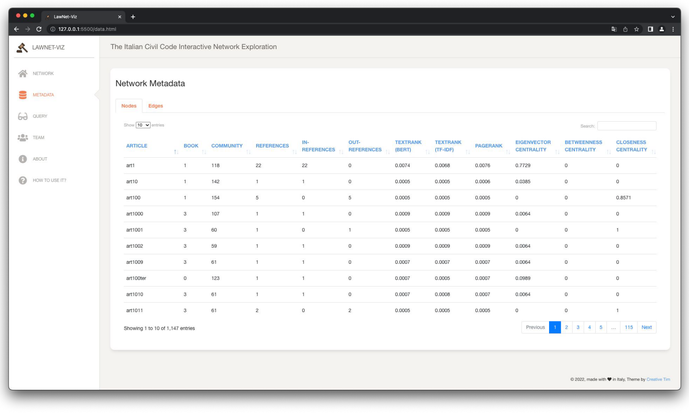
1. NETWORK MODE



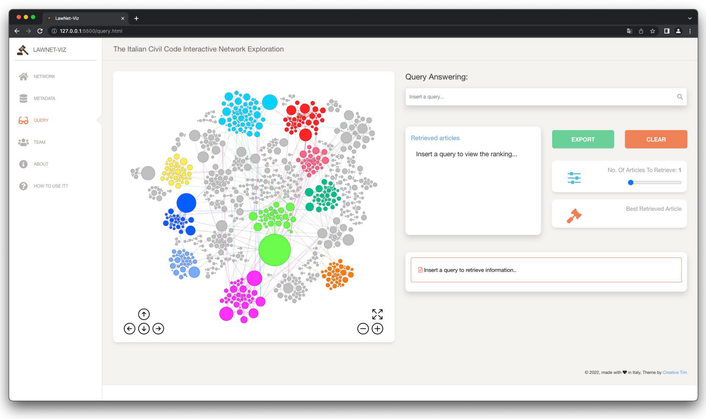
- User-friendly exploration and manipulation of the article references network
- Articles' neighborhood inspection
- Topological and content statistics

2. METADATA MODE

- Interactive tabular exploration of the metadata associated with articles and references between articles
- Data filtering and sorting based on user needs



3. QUERY MODE



- Retrieval of articles relevant to a given input natural language query
- Selection of text representation model
- Reference-network positioning and content inspection of the results

CASE STUDY

- The Italian Civil Code (ICC), the legislation source containing norms that regulate private law in Italy
- Integration with the LamBERTa framework, a BERT-based framework which models law article retrieval as a prediction problem, finetuned on the ICC

PATH TO IMPACT

Enhance access to justice to legal professionals, government and administrative agencies:
- Reduce the workload
- Improve productivity
- Ensure efficiency and effectiveness

Shrink the gap-to-justice involving citizens and laymen:
- Act as touch-point with a potentially intricate domain
- Simplify access to regulatory systems

LawNet-Viz is currently planned for product development, with initial focus on the landscape of Italian and EU legal sources.

L. La Cava, A. Simeri, A. Tagarelli. **The Italian civil code network analysis**. Proc. RELATED – Relations in the Legal Domain Workshop @ICAIL 2021. CEUR Workshop Proceedings, 2021, 2896, pp. 3–16

A. Tagarelli, A. Simeri. **Unsupervised law article mining based on deep pre-trained language representation models with application to the Italian civil code.** Artificial Intelligence and Law, 2021. DOI: 10.1007/s10506-021-09301-8

MORE INFORMATION: